

V4.2 (10 main actions)

- Sense

- Retrieval Actions

This one is special, it only has main actions

Retrieve from skill library

Subaction	Examples & Quotes
Retrieve from skill library	Grab pre-built skill snippet (Minecraft “chop tree”) <i>Find and load ready-made code snippet from pre-built library</i>

Retrieve from local corpus

Subaction	Examples & Quotes
Retrieve from local corpus	<i>“diverse filetype reading; read between 1 to 15 documents and/or tables” (P2)</i> <i>"read the README file — Phase 1: read_file('reproduction_package/readme.txt')"</i> (P11)

Retrieve from external knowledge base

Subaction	Examples & Quotes
Retrieve from external knowledge base	<i>"LLM which generates buggy code by integrating retrieved error slices from BugRAG as context" (P29)</i> <i>"queries BugRAG to check for existing entries" (P29)</i> <i>Recall three relevant distinct problems (P28): "Recall three (03) relevant and distinct problems (different from the user task)" (P28)</i>

Retrieve from open web

Subaction	Examples & Quotes
Retrieve from open web	<i>"high reliance on web search to offload knowledge retrieval to external sources" (P16)</i>

Retrieve relevant context

Subaction	Examples & Quotes
Retrieve relevant context	<i>"LLM leverages retrieved HLS-related context to transform input C algorithms or natural language descriptions" (P29)</i>

- Memory Actions

Store Information

Subaction	Examples & Quotes
Store information in working memory	Scratchpad for current inputs and intermediate results <i>Quick-access buffer that holds recent inputs and intermediate results</i>
Store episodic trajectories	Record full action sequences for later training <i>Save complete sequences from start to finish for training/review</i>
Store knowledge in semantic memory	Save general world facts <i>Save general facts about the world, not tied to any specific event</i>
Store experiences in episodic memory	Save events as episodes <i>Saves past events as personal-diary-like episodes (which game won, which plan failed)</i>
Store information in long-term memory	Store information in external storage <i>"Gemini-3-Pro...initialize_memory_bank (7 times) "Gemini-3-Pro...update_memory_bank (22 times)...attempts to persist state and key information externally"" (P16)</i> Maintain curriculum library <i>Keep a syllabus of mastered and upcoming skills, arranged by difficulty, so the agent learns in a sensible order.</i>

Update Information

Subaction	Examples & Quotes
Update memory	Update experiences <i>refinement of memories (represented in text) as the system encounters new scenarios"</i>

Discard information

Subaction	Examples & Quotes
Discard information from working memory	Discard local memory after subtask completion (P1) <i>"the local memory is discarded upon successful completion of the subtask" (P1)</i>
Discard redundant game state, keep summaries (P25)	<i>"automatic context compaction to manage thousands of reasoning steps, preserving only LLM responses and action summaries while discarding redundant game state" (P25)</i>

Consolidate memory

Subaction	Examples & Quotes
Consolidate working memory into long-term memory	Retain only successful analysis path to global memory (P1) <i>This discard-after-use strategy ensures that only the clear and successful analysis path is retained in the global memory, preventing interference from the intermediate trial-and-error history and thereby maintaining a concise and efficient context for subsequent steps (P1)</i>

Compact context window	<i>"automatic context compaction to manage the thousands of reasoning steps required, preserving only LLM responses and action summaries while discarding redundant game state" (P25)</i>
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Read memory

Subaction	Examples & Quotes
Read from working memory	Grab intermediate results from scratchpad <i>Check the scratchpad — grab stored intermediate results and state</i> “IMPLICITMEMBENCH reframes evaluation from ‘what agents recall’ to ‘what they automatically enact’.” (P12)
Read from long-term memory	Importance-weighted retrieval of discoveries (P25) <i>“Persistent memory system storing discoveries (locations, NPCs, items, strategies) with importance-weighted retrieval” (P25)</i>

● Think

● Planning Actions

Decompose task

Subaction	Examples & Quotes
Decompose into subtasks	<i>"LLM-based planner to decompose the specified language instruction L into a sequence of subtasks $S = [S0, S1, ..., Sn]$" (P34)</i> <i>"Decomposing hard problems into subproblems often makes them easier and more efficient to solve" (P17)</i> <i>"Decomposes complex user requests into manageable subtasks, an Executor that carries out the analysis by generating and running code, and an Evaluator that assesses the quality of the results" (P1)</i> <i>"formulate plans, decompose problems into sub-steps" (P2)</i> <i>todo_write tool (15 invocations by GPT-5.2) (P16)</i>
Decompose into subgoals with success conditions	<i>"LLM decomposes the task into a sequence of subgoals, each paired with an executable success-condition function" (P25)</i>
Decompose by role specialization	<i>"splitting the instruction-following challenge into distinct reasoning and grounding roles handled by a reasoner agent and an actor agent" (P34);</i> <i>"Specialists each solve a distinct task type, receiving only the relevant portion of the input" (P17)</i>

Formulate a workflow or plan

Subaction	Examples & Quotes
Formulate a high-level plan	<i>"METRO starts with the identification of the strategic action a_i for every expert utterance u_i in transcript D" (P10);</i> <i>"summarizing it into a high-level planning directive that emphasizes cumulative temporal effects" (P8)</i>

Formulate an analysis workflow	<i>"the Planner accurately interprets user intent and formulates a comprehensive analysis workflow" (P1)</i>
Plan navigation through environment	<i>"planning to navigate the web; tasks require navigation through an average of 4.2 web pages" (P2)</i>
Plan function or tool use	<i>"I can use the pandas method corr() to calculate the Pearson correlation coefficient" (P32)</i>
Plan Code or artifact structure	<i>"Planning: HTML Structure, CSS Styling, JavaScript Functionality" (P28)</i>
Formulate plan from template	<i>"HLSTuner formulates a detailed plan that specifies: (1) the combination of HLS directives, (2) target code segments, and (3) the insertion actions" (P29)</i>

Select Strategy

Subaction	Examples & Quotes
Select among candidate strategies	<i>"selects effective HLS directive combination strategies and inserts directives within the specific structure (e.g., loops and arrays)" (P29)</i>
Switch to fallback strategy	<i>"incorporating a dummy score prediction as a fallback mechanism" (P11)</i>

Modify Plan

Subaction	Examples & Quotes
Replan dynamically based on feedback	<i>"RequireReplan provides the LLM-based executor with the capability to dynamically adjust the plan" (P34)</i>
Refine Requirements	<i>"Refine requirement based on validated test cases to ensure alignment" (P28)</i>

● Reasoning Actions

Generating

Subaction	Examples & Quotes
Generate candidate options	Brainstorm possible next moves <i>Generate one or more possible next moves</i> <i>"Running multiple candidate pipelines for each analysis step" (P1)</i> <i>"generates solutions based on analogies to unrelated projects, which resemble few-shot prompting" (P28)</i> <i>"Fork the current generative state and generate a new parallel trajectory from this point of high uncertainty" P4</i> <i>"the LLM reinterprets the actions from breadth logic...summarizing them into a concise next-step strategy" (P8)</i>

Generate structured artifacts	<i>"Each agent then emits a set of initial codes; cluster semantically similar codes and generate preliminary themes" (P27);</i> <i>"automatically generates candidate user-facing requirements" (P28);</i> <i>"Create JavaScript functionality handling empty display and consecutive operator clicks" (P28)</i> <i>"break down the given character description into multiple atomic pieces of knowledge" (P22)</i>
Generate evaluations	<i>"LLM Group for multifaceted evaluation generates diverse debugging instructions" (P29)</i> <i>"For every binary score $z(q)_{i,j}$ from the judge, there is a corresponding explanation $e(q)_{i,j}$" (P3)</i>

Analysing

Subaction	Examples & Quotes
Analyse artifact structure and behavior	<i>"Analyze frontend framework and interactive components" (P28)</i> <i>"analyzes the project's core functionalities and their interactions with UI elements by reading the source code" P28</i>
Detect patterns or trends in data	<i>"Identifying patterns, directions, or changes in data over time or across contexts" (P2)</i> <i>"Measuring relationships or associations between two or more variables" (P2)</i>
Interpret meaning of artifacts	<i>"logical reasoning to interpret papers and code; mathematical reasoning to modify and run code; causal reasoning to infer scientific insights from results" (P11)</i>

Classify inputs into categories	<i>delegator can determine which of the k tasks the input string belongs to by observing only a constant number of metadata tokens (P17)</i>
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Explaining

Subaction	Examples & Quotes
Explain reasoning or outcomes	<i>"Explain the failure from a user requirement perspective" (P16)</i>

Summarizing/Distilling

Subaction	Examples & Quotes
Summarize recent observations and trajectories	Condense recent observations and trajectories into key takeaways

Inferring

Subaction	Examples & Quotes
Infer hidden state from observable evidence	Damage calculation reveals stat distributions (P25) <i>"intelligently infers hidden information through game mechanics: damage calculations reveal stat distributions, move priority ordering constrains speed ranges" (P25)</i>

Infer causal relationship	Determine plausible causal factor (P2) <i>“determine the most plausible event or factor accounting for this variability” (P2)</i>
Infer structure from indirect evidence	<i>"agents must infer the directory layout by inspecting package structures and README files" (P11)</i>
Infer errors	<i>M_q has the capacity to identify exactly one bug at a time (P17)</i> <i>“The analysis LLM then examines the error causes and provides debugging instructions” (P29)</i>

Comparing & Ranking

Subaction	Examples & Quotes
Compare values across sources	Quantify and compare across sources (P2) <i>“Quantifying occurrences and comparing values across sources or categories” (P2)</i>
Rank items by criteria	Order items by importance (P2) <i>“Ordering items or facts based on specific criteria or importance” (P2)</i>

Contextualizing

Subaction	Examples & Quotes
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Package prior reasoning as context for subsequent calls	<i>Feed reasoning so far as context to next LLM call</i>
Construct structured context object	<i>“global interactive multi-behavior overlapping network is constructed based on the simulated behaviors of reposting and reposting with a comment, as well as following” (P21) “construct prompt templates for creating reader agents based on user attributes ua and user historical posts” (P21)</i>
Configure agent persona or role-conditioning	<i>“I want you to play as {role}, imitating {role}’s personality and values” (P22)</i> <i>“maintain its role-playing ability even when refusing to answer” (P22)</i> <i>“A pool of k=4 role-conditioned GPT-4o agents are each given the full interview transcript as input” (P27)</i> <i>“others may retain role-specific perspectives to support diverse theme formulation” (P27)</i>
Assign roles in a multi-agent team	<i>specifying a scope, or role, for an agent allows it to be treated as a computational object. map each on to an employee in a human organization (P17)</i> <i>"assembling a team of three LLM roles — analyst, coder, and tester — responsible for analysis, coding, and testing" P28</i> <i>"assigns diverse roles to multiple agents to efficiently decompose complex tasks" (P28)</i>

Combining

Subaction	Examples & Quotes
Combine information from multiple sources	<i>“combine information from multiple sources... to produce a coherent solution” (P2)</i>

	<i>"the final results from all subtasks are synthesized to meet the user's requirements" (P1);</i> <i>"Determining a representative value that summarises numerical data collected from multiple sources" (P2)</i>
Aggregate observations into a structured representation	<i>"supplementary semantic map M't, which aggregates the information from Mt over successive time steps. The intuition resembles a form of majority voting" (P34)</i> <i>"aggregates prediction results from multiple tools to generate the final cell type labels" (P1)</i>

Filtering

Subaction	Examples & Quotes
Filter information by threshold	<i>"Selecting relevant information based on criteria, quality, or thresholds" (P2)</i>

● Evaluate Actions

Evaluating with gold

Subaction	Examples & Quotes
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Compare against gold reference	<i>“we prompt LLM to review the correct HLS-C code, pairing buggy code segments with corresponding error messages to construct debugging CoT” (P29)</i> <i>“HLSFixer retests the corrected HLS design against the golden results to ensure semantic equivalence” (P29)</i>
Score on gold criteria	<i>“The Gold Checker evaluates each annotation on three binary criteria: equivalence, completeness, and correctness, where a score of 1 indicates the criterion is met” (P28)</i>

Evaluating with goals/requirements/constraints

Subaction	Examples & Quotes
goal-completion check	<i>“Independently checks whether objectives are truly complete, preventing the orchestrator from advancing when the main agent incorrectly believes a task is finished” (P25)</i>
Requirement-satisfaction check	<i>“the final results from all subtasks are synthesized to meet the user’s requirements” (P1)</i>
Constraint / budget check	<i>“HLSTuner enables QoR-aware reasoning to align optimization goals with hardware constraints” (P29)</i> <i>“if hardware utilization exceeds the budget” (P29)</i>
Domain-rule / best-practice check	<i>“codified best-practices, such as the standard order of operations (e.g., quality control must precede normalization)” (P1)</i> <i>“Agents share project context to ensure: Cross-file consistency; Non-conflicting test-ids” (P28)</i> <i>“Detect logical errors, missing functionalities, or violations of best practices” (P16)</i>

Evaluating without ground truth

Subaction	Examples & Quotes
Score on quality dimensions	<i>"we use GPT-4o as the default evaluator, each dimension scored on a scale of 0 to 2" (P22)</i> <i>"map the agent's deliverables to a score ranging from 0 to 10" (P16)</i> <i>"Evaluation Scores: Obtain $s^*(t) = (C, D, T)$ on the trustworthiness dimensions" (P27)</i> <i>"the scoring agent (functioning as the judge) based on clarity, logical soundness, alignment with error messages, this agent selects the optimal suggestion" (P29)</i>
Check rubric compliance	<i>"Verify the implementation of every item listed in {Rubrics}" (P16)</i> <i>"list exactly which rubrics were not met" (P16)</i>
Evaluate visual/behavioral correctness	<i>"Evaluates dynamic behavior and visual correctness based on screenshots" (P16)</i> <i>"Analyze {Visual Assets} for UI elements, layout consistency, and text rendering" (P16)</i>
Evaluate internal consistency	<i>"task an evaluator agent with determining whether each theme is consistent with its supporting quotes" (P27)</i> <i>"In Phase 2, they inspect the provided code for potential inconsistencies; generating a reproducibility score on a scale from 1 to 4" (P11)</i>
Evaluate intermediate results	<i>"agents with domain expertise verify intermediate results and reduce errors" (P28)</i>
Provide qualitative judgement	<i>"Upon successful generation of a result, the Evaluator agent, ALLMe assesses the outcome and provides natural language feedback if it deems revisions are necessary." P1</i>
Simulate counterfactual outcomes	Mentally simulate an alternative scenario — either before acting (imagine 'if I do X, what happens?') or post-hoc (imagine 'what would have happened if action X had been different?'). Includes counterfactual trajectory generation for causal analysis. (P33) <i>"analyzing this effect necessitates counterfactual reasoning across three distinct scenarios"</i>

Validate predicted issues	<i>“The agent assesses the contextual applicability of potential bugs, reducing the probability that the LLM forcibly generates trivial results” (P29)</i>
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● Deciding Actions

Make a decision

Subaction	Examples & Quotes
Make a decision according to memory	<i>The agent conditions its decisions on I, the current observation st, and a history buffer of n past state-action pairs $P5$</i>

Pick scores

Subaction	Examples & Quotes
Select action by score	<i>argmax = highest score; softmax = probabilistic sample; majority vote across evaluators</i>

Decide accept or not

Subaction	Examples & Quotes
Decline out-of-scope queries	<i>“appropriately reject queries that exceed their knowledge boundaries or conflict with their role settings” (P22)</i>

Decide under uncertainty

Subaction	Examples & Quotes
Fork trajectory at uncertainty	<i>"Fork the current generative state and generate a new parallel trajectory from this point of high uncertainty" (P4)</i>

● Act

● Grounding Actions

How agent interacts with external world

Interact with users

Interact with users: Actions where the agent communicates with human users: receiving inputs from them, requesting their input, conveying outputs to them, engaging socially, or self-regulating its behavior in their presence.

Subaction	Examples & Quotes
Accept instructions from humans	Receive task or command from human user <i>e.g. "book me a flight" or "translate this paragraph"</i> <i>"agent receives task queries and deliverables, completing tasks through multi-turn interactions" (P16)</i> <i>"demonstrate clear intent understanding, partial progress" (P5)</i>
Ask for clarification from people	Proactively ask human when unclear <i>e.g. "Which file did you mean?"</i>

Communicate task outcome natural language	<i>Agent informs the user: 'Buy a nice rich navy bathing dress' (P3)</i> <i>Agent informs the user that wifi is turned off (P3)</i>
Communicate via visualization	<i>"CellAgent generated differential expression and marker gene visualizations, enabling intuitive interpretation of cluster identities" (P1)</i>
Communicate via structured format	<i>"generate outputs in a JSON format (or lists of JSON objects)" (P2)</i>
Communicate refusal/inability	<i>"appropriately reject queries that exceed their knowledge boundaries or conflict with their role settings" (P22)</i> <i>recognize and refuse queries that conflict with their role knowledge" (P22)</i> <i>"providing clear refusal responses with appropriate explanations" (P22)</i>
Express tone to user	<i>"models exhibit a strong default tendency to excessively affirm, apologize, and express gratitude" (P11)</i>
Disclose self-information to user	<i>"identity disclosure, S2S systems often proactively mention that they are intelligent assistants (P11)</i>

Interact with physical environments

Subaction	Examples & Quotes
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Perceive physical environment	Convert images to text via VLMs <i>VLM image-to-text description</i> Convert sensor / image / audio readings to text <i>Turn images, sounds, or sensor readings into text descriptions for text-based LLMs</i> <i>“Explore skill enhances the LLM-based executor’s ability to guide the agent in room exploration by sampling navigation goals from traversable areas ” (P33)</i> <i>Covert perceptual visual information to map</i> <i>“semantic mapping module receives egocentric visual observations...processed into a depth map and instance segmentation using a UNet and a MaskRCNN” (P34)</i>
Affect physical environments	Send language commands to robot arm <i>Move things in the real world by sending language commands to a robot arm</i> <i>“the action Play is utilized to interact with the environment or request re-planning of the current plan S” (P34)</i>

Interact with digital environments:

Actions through which the agent navigates digital interfaces, modifies the state of digital objects, or issues operational commands within software systems.

Boundary with retrieval: Read-only information retrieval from digital sources is coded under Retrieval Actions; computational tool invocations (e.g., code execution, calculators) are coded under Augment with external computation.

Subaction	Examples & Quotes
Navigate digital interfaces	Browse websites, scroll pages, navigate menus, move within apps

	<i>"navigate multiple websites, extract information from both structured and unstructured sources" (P2)</i>
Modify digital objects	Annotate UI Components Add data-testid attributes (P28) <i>"annotate_interactive_components(file, strategy='add data-testid')"</i> (P28)
Issue operational commands	Invoke structured API or data endpoint <i>"Game Control: get_game_state, press_buttons, and navigate_to for direct game control"</i> (P25)

Interact with other agents

Subaction	Examples & Quotes
Monitor peer agent's state or decision	Observe another agent's chosen action before making one's own decision — a prerequisite for oversight, override, or trust-based deference in multi-agent systems. (P33) <i>"observes both S_0 and A_0, and decides whether to override the AI's treatment or not"</i>
Receive feedback from peer agent	<i>Evaluator provides natural-language feedback (P1)</i>
Send message to peer agent	<i>"agent engages in multi-turn interactions...to generate deliverables" (P16)</i>

Recommend action to peer agent	Communicate a proposed action to a peer agent for their consideration, knowing the peer may accept or override it — agent-to-agent advisory communication. (P33) <i>"the AI recommends one of eight possible treatments, which is then reviewed and potentially overridden by the clinician"</i>
Override peer agent's decision	Replace another agent's selected action with one's own decision — the active side of human-in-the-loop oversight or hierarchical agent control. (P33) <i>"If the clinician overrides the AI, the patient outcome Y is determined by S_0 and the alternative treatment H_0 suggested by the clinician"</i>
Dispatch task to sub-agent	<i>"A central orchestrator maintains a high-level route plan while dynamically dispatching sub-agents based on game context" (P25)</i>
Argue or debate with peer agent	Multi-agent argumentation toward better answer <i>Have multiple agents argue different sides to reach a better answer</i>

Augment with external computation:

Subaction	Examples & Quotes
Execute code	<i>Test Runner detects logic inconsistencies (P28)</i> <i>Test Runner executes the script to detect logical inconsistencies or failed assertions. (P28)</i>
Invoke specialized computation tool	Use calculator / heavy compute tools <i>Call outside tools to handle heavy math</i> <i>run_shell_command tool (425 and 362 invocations, respectively) (P16)</i> <i>"necessitates over 90 precise tool calls, significantly raising the bar" (P16)</i>

	<i>"deliverables are synced to a Docker-based remote sandbox...emulates human computer operations" (P16)</i>
Invoke visual inspection tool	Zoom-in tool for region-of-interest (P4) <i>"proactively invokes the Zoom-in tool for a targeted examination of the specific region of interest" (P4)</i>

● Executing

Executing plan

Subaction	Examples & Quotes
Execute strategy	<i>"an insertion agent executes this plan for HLS-C optimization" (P29)</i>

Executing debug

Subaction	Examples & Quotes
Adopt debugging instructions	<i>"Operating under strict instruction adherence, this agent adopts the instructions to implement debugging" (P29)</i>
Rewrite code after bug fix	<i>debugging specialist rewrites the entire code every time it fixes a bug (P17)</i>

Terminating

Subaction	Examples & Quotes
Provide final answer	<i>"Final Answer: The average number of lines per scene in the ep_7.csv dataset is approximately 6.02" (p32)</i> <i>"produces a final answer and terminates the rollout; enclose it within <answer></answer> tags" (P4)</i>
Generate refusal	<i>"providing clear refusal responses with appropriate explanations" (P22)</i>

● Adapt

● Learning Actions

Learning reasoning

Subaction	Examples & Quotes
Update reasoning via prompt update	Rewrite own prompt template <i>Learn a better way to reason via prompt rewrite</i>

Learning grounding

Subaction	Examples & Quotes
Update grounding via code-based skills	Improve web-navigation code snippets <i>Write or improve code that interacts with outside world</i>

Update retrieval procedures	Improve how the agent searches for information
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Learning knowledge

Subaction	Examples & Quotes
Update source code as procedural memory	Self-patch agent's own source code <i>Modify the agent's own source code to change behavior</i>
Update semantic memory with knowledge	Expand Error Repository with new mnemonic (P29) <i>"the inspection agent identifies a new error type and integrates the slice into the error repository with a new mnemonic identifier" (P29)</i>
Update memory from new experiences	<i>refinement of memories (represented in text) as the system encounters new scenarios (P17)</i>
Update hypothesis with new evidence	<i>"allowing the model to update its textual understanding, refine its hypothesis, or even trigger further visual exploration" (P4)</i>

Learning LLM parameters

Subaction	Examples & Quotes
Update parametric policy	Change the model's internal weights through training <i>Real learning by changing internal weights</i>

Update LLM parameters via SL/RL/RLHF	SL/RL/RLHF weight adjustment <i>Use supervised, reinforcement, or human-feedback learning</i>
Update action parameters based on feedback	Auto-adjust params from exception info (P1) <i>“automatically adjusts parameters using exception information to generate executable code” (P1)</i>

Learning instructions

Subaction	Examples & Quotes
Learning instructions	Extract rule from examples for future use <i>Look at input→output examples, figure out underlying rule</i>

● Reflection Actions

Reflect on errors and failures

Subaction	Examples & Quotes
Diagnose failure against ground truth	<i>“Analyzes stuck states by comparing current situation against ground truth sources (porymap data, knowledge base) to diagnose navigation failures” (P25)</i>
Inspect Error Pattern	<i>“an inspection agent examines the erroneous code and error messages parsed from the HLS tool test results” (P29)</i>

Analyze log to formulate fix instructions	Reasoning-to-instruction on HLS log (P29) <i>"an analysis agent adopts a reasoning-to-instruction method, analyzing the HLS log to formulate error modification actions" (P29)</i> <i>"This model formulates explicit modification instructions with detailed analysis" (P29);</i> <i>"Provide explicit instructions on what must be rectified in the next iteration" (P16)</i>
Reflect on failed episodes for prior knowledge	<i>"RequireReplan...dynamically adjust the plan, improving the robustness to exceptions" (P26)</i> <i>"the grounded prior knowledge prevents the agents from repetitive errors and facilitates grounded exception handling" (P34)</i> <i>"This allows the agent to learn from mistakes in realtime and avoid repeating errors before the local memory is discarded" (P1)</i>
Self-Correct Step Implementation	<i>"Modify step implementation based on error message" (P28)</i> <i>"In case of an execution error E(ci), the Executor autonomously performs self-correction to produce a valid code version" (P1)</i> <i>"When the initial attempt fails... HLSTuner activates an iterative refinement incorporating current directives and the resulting QoR" (P29)</i>

Reflect on self-outcomes

Subaction	Examples & Quotes
Self-reflect	<i>The results of the execution are fed back, enabling GPT-4 to refine its responses and propose further iterations" (P32)</i> <i>"This iterative cycle continues until GPT-4 determines that the accumulated information suffices to conclusively answer the problem" P32</i> <i>"This self-reflective optimization loop iterates to enhance precision, and the final results from all subtasks are synthesized" (P1)</i>

	<i>"the optimization process runs three iterations, each invoking a different algorithm" (P1)</i>
Reflect on proposed fix	<i>"reflects on the code after assuming the fix to ensure the modification is reasonable" (P29)</i>
Pre-action self-check	<i>"These steps are labeled as reflective, indicating the model is engaging in self-monitoring or explicit reasoning before issuing commands" (P1)</i>
Refine Strategy Iteratively	Iterative refinement using current QoR (P29) <i>"When the initial attempt fails... HLSTuner activates an iterative refinement incorporating current directives and the resulting QoR" (P29)</i>

Reflect on external feedback

Subaction	Examples & Quotes
Receive and integrate external feedback	<i>"the Evaluator agent ALLMe assesses the outcome and provides natural language feedback if it deems revisions are necessary" (P1)</i> <i>"feedback from the feedback agent is used to iteratively refine and improve the generated themes" (P27)</i> <i>"leverage feedback to markedly improve performance...feedback-driven self-correction" (P16)</i> <i>The Evaluator drives a self-reflective optimization mechanism, which leverages automated evaluation methods for various analysis tasks to iteratively refine outcomes, replacing subjective manual assessments. P1</i>